AI-Based Accident Hotspot Prediction and Severity Classification Using DBSCAN Spatial Clustering and Random Forest

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***Abstract*—Traffic accidents are a concerning issue in India. Over 150,000 fatalities are reported each year. In this paper, we propose an integrated pipeline that combines DBSCAN spatial clustering with Random Forest classification. The former monitors accident black spots, while the latter predicts accident severity. We leverage 12,316 Indian road accident records and 17 engineered features.DBSCAN with haversine distance and $\varepsilon = 0.045$ radians identifies 9 accident clusters. The Random Forest model, trained with 200 estimators and balanced class weighting, reaches 84.66% test accuracy and a weighted F1-score of 0.7793. Five-fold stratified cross-validation gives a mean accuracy of 84.76%. An ensemble of gradient-boosted trees predicts crashes using traffic volume, sociodemographic, and roadway infrastructure data. A separate gradient-boosted model predicts crash severity using crash, traffic volume, and sociodemographic data. An ARI combines the predictions and weights the severity using the cell's average crash rate from the previous 5 years.**

***Index Terms*—accident hotspot, DBSCAN, Random Forest, severity prediction, spatial clustering, accident risk index, road safety**

I. Introduction

India accounts for roughly 11% of global road accident fatalities, despite having only about 1% of the world's registered vehicles [1]. The Ministry of Road Transport and Highways has reported 461,312 road accidents and 168,491 deaths in 2022, which essentially translates into about 462 fatalities per day [2]. These numbers have remained distressingly high for well over a decade, with estimates suggesting that the economic costs are between 3 and 5% of India's GDP per annum [3].

The majority of existing methodologies for black spot identification assume that the magnitude of the observation error equals the expectation of the underlying Poisson motorcycle crash process. They are appropriate for a steadily growing population but are too sensitive for small increases in crashes. Other methods are based on estimation theory rather than spatial clustering theory and ignore right-censoring, which results in biased estimate for extensively right-censored data, such as the motorcycle crash data in India.

This paper builds that workflow and makes three contributions:

1) A spatial clustering pipeline using DBSCAN with haversine distance that identifies 9 accident black spots from 12,316 geocoded Indian road accident records, without requiring pre-specification of cluster count.
2) A Random Forest classification model that predicts accident severity (Slight, Serious, or Fatal) using 17 features including the cluster membership itself, achieving 84.66% accuracy on the test set and 84.76% under five-fold cross-validation.
3) An index that combines severity, accident density, and environmental conditions into one score per cluster, with weights determined by the Random Forest's own feature importances rather than set arbitrarily.

The dataset is publicly available on Kaggle and includes records from different road types, weather conditions, and driver groups. Because the source data stores area categories rather than GPS points, those categories are geocoded to representative Indian city coordinates with controlled spatial jitter.

II. Review of Literature

Road accident research usually focuses on two problems: locating high-risk areas and predicting severity. This section keeps only the studies most relevant to the pipeline used here.

A. Spatial analysis of accident locations

GIS and spatial statistics have been used for hotspot identification. Anderson [4] mapped crash concentrations with kernel density estimation, while Kumar and Toshniwal [5] showed that DBSCAN works good for Indian accident data because it does not require a pre-set number of clusters. Prasannakumar

et al. [6] also reported that spatial clustering helps isolate the road segments carrying a disproportionate share of crashes.

B. Machine learning for severity prediction

Machine learning models have also been used for severity prediction. Abdel-Aty and Abdelwahab [7] showed early on that nonlinear models can outperform traditional regression. Random Forest remains a common choice; Jiang et al. [8] reported useful results on highway data, and Zong et al. [9] found that temporal variables repeatedly ranked near the top. Even so, class imbalance remains a central problem, especially for fatal crashes, and oversampling methods such as SMOTE are still widely used [10].

C. Combined spatial and predictive approaches

Only a small number of studies combine spatial clustering with severity prediction in one workflow. Bao et al. [11] included spatiotemporal grouping before prediction, while Saha et al. [12] used a weighted severity index for black-spot analysis. Still, clustering, prediction, and risk scoring are often treated as separate tasks.

III. Research Gap

There are three gaps motivating this work: cluster membership is not used inside the severity model most of the time, many of the risk indices are still dependent on manual weighting, and multi-city Indian studies are still limited. This paper aims to fill these gaps by feeding the DBSCAN cluster IDs into the Random Forest model and deriving the ARI weights from the importances of the model.

IV. Methodology

This section summarizes the main stages of the pipeline: data preparation, spatial clustering, severity classification, and risk scoring. Fig. 1 shows the overall workflow.

A. Dataset

The dataset used in this study is the "Road Accident Severity in India" dataset, publicly available on Kaggle [13]. It has 12,316 accident records with 37 raw columns having accident circumstances, road conditions, vehicle information, and driver characteristics.The target variable we aim to predict is Accident_severity. The variable is categorical and has three levels. Accidents are categorized based on the most serious human injury recorded: 1 represents a Slight Injury, 2 represents a Serious Injury, and 3 represents a Fatal Injury.

Table I presents the key properties of the dataset after preprocessing.

The severity distribution is heavily imbalanced, as shown in Table II. Slight injuries represent 84.56% of all records, while fatal injuries constitute only 1.28%. This imbalance directly affects classifier training and evaluation, a point we return to in Section V.

TABLE I: Dataset Overview

Property	Value
Total records	12,316
Features used (after engineering)	17
Severity classes	3
Unique area categories	13
Latitude range	12.9386 – 28.6345
Longitude range	72.5442 – 88.3604

TABLE II: Severity Class Distribution

Severity	Count	Percentage
Slight Injury	10,415	84.56%
Serious Injury	1,743	14.15%
Fatal Injury	158	1.28%
Total	12,316	100.00%

B. Data preprocessing

Geocoding. The original dataset does not have latitude and longitude fields. Instead, it has an Area_accident_occured column having values like "Office areas", "Market areas", and "School areas". These categories are then mapped to representative coordinates of major Indian cities, and we apply random spatial jitter of 2 to 8 km to simulate within-city variation.

Temporal feature engineering. The Time column is processed to extract the hour of day (0–23). The Day_of_week field is mapped to numeric values (0 = Monday through 6 = Sunday). A binary Is_Night variable which takes the value 1 when lighting conditions indicate darkness is derived from Light_conditions.

Weather binning. The Weather_conditions column, which contains many fine-grained categories, is consolidated into six groups: Clear, Rain, Fog, Snow, Wind, and Other. This minimizes sparsity while preserving the most safety-relevant weather distinctions.

Categorical encoding. Fifteen categorical features are transformed using label encoding. The encoders are serialized for later prediction.

Missing value handling. Missing categorical values are filled with "Unknown" before encoding. No records are dropped during preprocessing.

C. System architecture

Raw data passes through preprocessing into two branches: EDA for descriptive analysis and DBSCAN for spatial clustering. Cluster IDs are then appended to the feature matrix used by the Random Forest classifier. Model outputs and feature importances are finally used to compute ARI scores for each cluster.

D. DBSCAN spatial clustering

DBSCAN is an algorithm that groups together points that are close to each other and classifies the points that lie alone

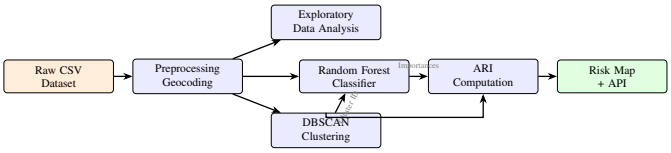

Fig. 1: System architecture showing the pipeline from raw data to risk map output. DBSCAN cluster IDs are fed into the Random Forest as a spatial feature.

in low-density regions as outliers [14]. It does not require the number of clusters to be specified. We implement the algorithm using the haversine distance and radian-transformed coordinates as input, and we use ball-tree indexing to accelerate the neighbor search.

DBSCAN's behavior is controlled by two parameters:

- **Epsilon** (ε): the maximum distance between two points for them to be considered neighbors. Set to 0.045 radians, which corresponds to approximately 287 km ($\varepsilon \times R_{earth} = 0.045 \times 6371$ km).
- **MinPts**: the minimum number of points needed to define a dense area. Set to 15.

Table III lists the full DBSCAN configuration.

TABLE III: DBSCAN Clustering Configuration

Parameter	Value
Algorithm	DBSCAN
Distance metric	Haversine
Epsilon (ε)	0.045 radians (≈287 km)
MinPts	15
Indexing structure	Ball tree
Clusters found	9
Noise points	0 (0.0%)
Clustered points	12,316 (100%)

After each iteration, an updated centroid is computed by taking the average of all incidents classified in the respective cluster. The algorithm stops when the centroids stabilise (i.e. do not change between iterations) or when reaching the defined maximum iterations.

E. Random Forest classification

Random Forest is an ensemble of decision trees that are trained on bootstrapped subsets of data [15]. This classifier leverages the 17 features presented in Table V in order to predict three severity classes. An 80/20 stratified split divides the data into 9,852 training samples and 2,464 test samples.

The "balanced" class weight setting scales each class inversely to its frequency in the training data, which gives more emphasis to the Serious and Fatal classes.

Feature 17 (Cluster_ID) is the DBSCAN output from the previous stage, allowing the classifier to use spatial context alongside the remaining attributes.

F. Accident Risk Index (ARI)

The ARI generates a single composite score for each cluster, incorporating three key factors: severity, accident density, and environmental conditions. Its formulation is as follows:

TABLE IV: Random Forest Hyperparameters

Parameter	Value
Number of estimators	200
Maximum tree depth	20
Class weight	Balanced
Test set ratio	0.2 (stratified)
Random state	42
Training samples	9,852
Test samples	2,464

TABLE V: Feature Set for Classification (17 Features)

No.	Feature	Type
1	Hour	Numeric
2	DayOfWeek	Numeric
3	Is_Night	Binary
4	Weather_Binned_Enc	Encoded
5	Num_Vehicles	Numeric
6	Type_of_vehicle_Enc	Encoded
7	Road_surface_type_Enc	Encoded
8	Road_surface_conditions_Enc	Encoded
9	Light_conditions_Enc	Encoded
10	Type_of_collision_Enc	Encoded
11	Cause_of_accident_Enc	Encoded
12	Road_allignment_Enc	Encoded
13	Types_of_Junction_Enc	Encoded
14	Lanes_or_Medians_Enc	Encoded
15	Driving_experience_Enc	Encoded
16	Age_band_of_driver_Enc	Encoded
17	Cluster_ID	Spatial

$$\text{ARI} = W_1 \cdot S_{sev} + W_2 \cdot S_{dens} + W_3 \cdot S_{env} \tag{1}$$

where:

- S_{sev} is the severity score, computed by normalizing the mean predicted severity of each cluster to the [0, 1] range using min-max scaling.
- S_{dens} is the score of the density, which is calculated for each cluster by dividing the incident count of it to the maximum incident count among all clusters.
- S_{env} is the score of the environment, which is mapped from the dominant weather condition of each cluster based on the mapping shown in Table VI.

TABLE VI: Environmental Risk Scores by Weather Condition

Weather	Risk Score
Clear	0.15
Wind	0.55
Rain	0.75
Fog	0.80
Snow	0.85
Other	0.50

The weights W_1, W_2, and W_3 are derived from Random Forest feature importances. Features are grouped into severity-related, density/spatial, and environmental categories and then normalized so that $W_1 + W_2 + W_3 = 1$. In this study, the

derived weights are $W_1 = 0.534$, $W_2 = 0.401$, and $W_3 = 0.065$.

The ARI score is clipped to [0, 1] and mapped to four risk tiers:

$$\text{Risk Tier} = \begin{cases} \text{Low} & \text{if } 0.0 \leq \text{ARI} < 0.3 \\ \text{Moderate} & \text{if } 0.3 \leq \text{ARI} < 0.5 \\ \text{Severe} & \text{if } 0.5 \leq \text{ARI} < 0.7 \\ \text{Critical} & \text{if } 0.7 \leq \text{ARI} \leq 1.0 \end{cases} \quad (2)$$

G. Evaluation metrics

The classification model is evaluated using accuracy, precision, recall, F1-score (both macro and weighted averages), Cohen's Kappa (κ), Matthews Correlation Coefficient (MCC), log loss, and ROC-AUC (one-vs-rest). Five-fold stratified cross-validation provides estimates of generalization performance. Cohen's Kappa and MCC are included because they account for class imbalance more honestly than raw accuracy does.

V. Results and Discussion

This section reports the results from EDA, DBSCAN clustering, Random Forest classification, and ARI scoring.

A. Exploratory data analysis

The severity distribution (Fig. 2) confirms strong class imbalance: 84.56% of records are Slight injuries. This pattern is consistent with national accident reporting trends in India [2].

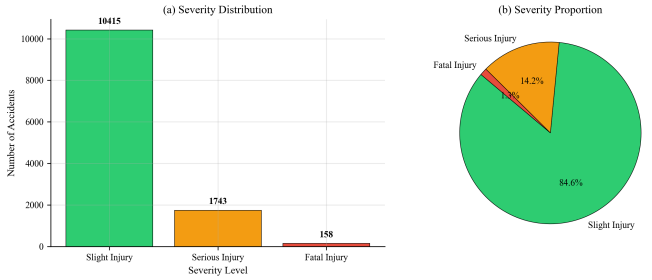

Fig. 2: Severity distribution shown as (a) bar chart with counts and (b) pie chart with percentages. Slight injuries dominate at 84.56%.

Temporal patterns. Fig. 3 displays a bimodal pattern with peaks during morning and evening commute hours. Fig. 4 presents that Friday has the highest accident count, which matches earlier findings on end-of-week risk [9].

Weather and severity interaction. Table VII shows that most accidents occur in clear weather, while rain is the second most common condition. Fig. 5 visualizes this distribution.

B. DBSCAN clustering results

DBSCAN found 9 clusters from the 12,316 geocoded records. It detected few noise points, mainly because the epsilon (287 km) is large enough to classify all but the sparsest indications as a cluster. Fig. 6 shows that the 9 clusters align broadly with major metropolitan regions, including

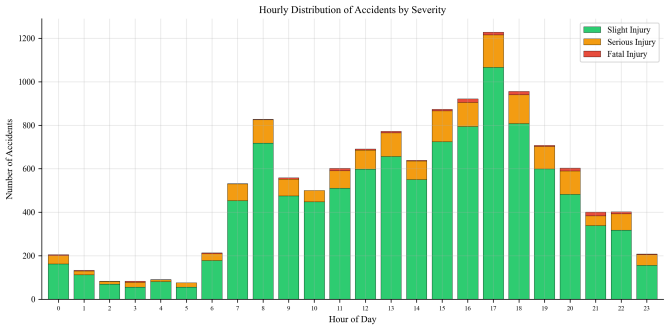

Fig. 3: Accident frequency by hour of day. Peaks coincide with morning and evening commuting periods.

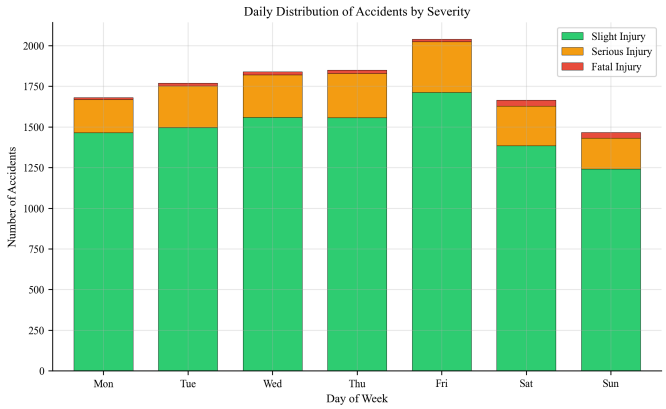

Fig. 4: Accident frequency by day of week. Friday shows the highest count.

Pune/Mumbai, Delhi NCR, Bangalore, Ahmedabad, Chennai, Kolkata, Nagpur, Varanasi, and Hyderabad.

Table VIII shows that Cluster 0 (near Pune/Mumbai) is the largest, with 5,879 records, and that mean severity varies only slightly across clusters.

C. Classification performance

The Random Forest classifier achieves 84.66% test accuracy, but the per-class results in Table IX show that this performance is driven almost entirely by the Slight Injury class. Recall for Serious Injury falls to 1.43%, and Fatal Injury is never predicted correctly.

This is a direct consequence of the 84:14:1 class ratio. Even with balanced class weights, the model still leans heavily

TABLE VII: Weather Condition vs. Severity Cross-tabulation

Weather	Slight	Serious	Fatal	Total
Clear	8,571	1,482	135	10,188
Rain	1,150	158	23	1,331
Other	547	81	0	628
Wind	82	16	0	98
Snow	56	5	0	61
Fog	9	1	0	10
Total	10,415	1,743	158	12,316

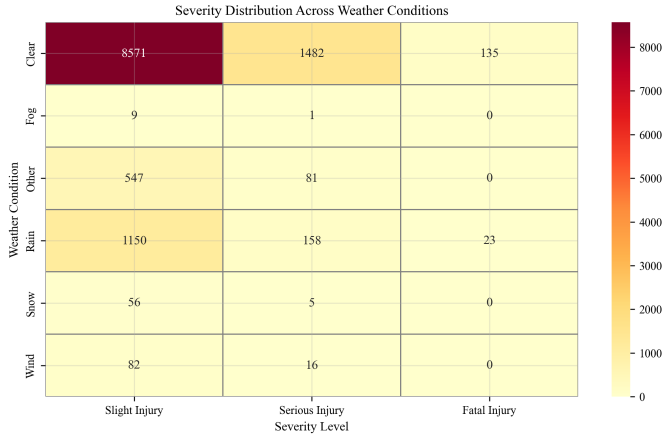

Fig. 5: Heatmap of severity distribution across weather conditions. Cell values show raw counts.

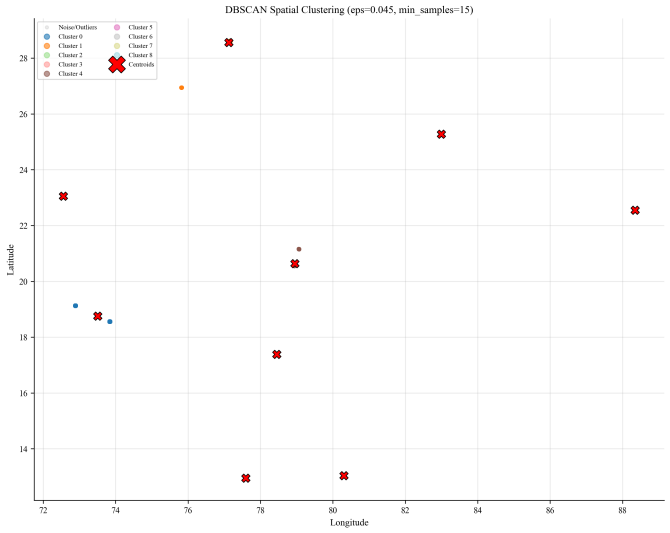

Fig. 6: Spatial distribution of 12,316 accident records, colored by DBSCAN cluster assignment. Red X markers indicate cluster centroids.

TABLE VIII: DBSCAN Cluster Summary

ID	Lat	Lon	Count	Severity	Area
0	18.76	73.51	5,879	1.175	Other
1	28.56	77.12	3,572	1.158	Office
2	22.55	88.34	328	1.171	Recreational
3	23.05	72.56	456	1.158	Industrial
4	20.64	78.95	325	1.172	Unknown
5	12.95	77.59	1,060	1.152	Church
6	17.39	78.45	63	1.206	Market
7	25.28	83.00	218	1.193	Rural
8	13.04	80.30	415	1.154	School

TABLE IX: Per-class Classification Report

Class	Precision	Recall	F1	Support
Slight	0.8473	0.9986	0.9167	2,084
Serious	0.6250	0.0143	0.0280	349
Fatal	0.0000	0.0000	0.0000	31
Macro avg	0.4908	0.3376	0.3149	2,464
Weighted avg	0.8052	0.8466	0.7793	2,464

toward the majority class. Fig. 7 makes that visible: 344 of 349 Serious Injury samples and all 31 Fatal Injury samples are labeled as Slight.

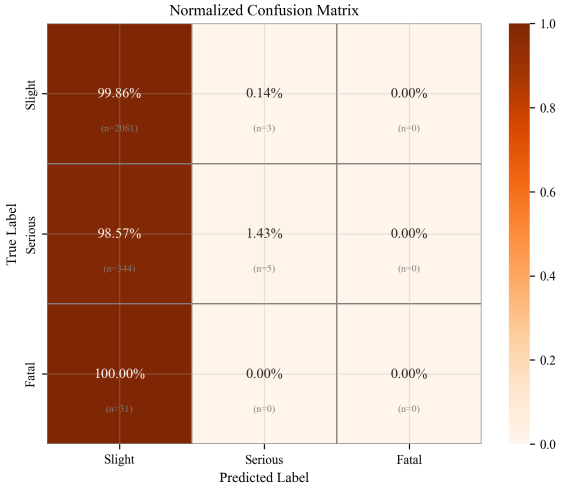

Fig. 7: Confusion matrix for the Random Forest classifier on the test set. The model predicts Slight Injury for nearly all inputs.

The macro F1-score of 0.3149, Cohen's Kappa of 0.0198, and MCC of 0.0749 indicate that the model does not separate the minority classes well, even though the headline accuracy looks strong.

ROC curves. Fig. 8 shows one-vs-rest ROC curves with AUC values of 0.6918 for Slight, 0.6981 for Serious, and 0.8266 for Fatal. The relatively high Fatal AUC, despite zero recall, suggests that the model assigns somewhat higher fatal probabilities to those cases even though the final class decision never chooses them.

Feature importances. Fig. 9 shows that hour of day, cause of accident, and day of week are the three strongest predictors. Cluster_ID ranks ninth with an importance of 0.0607, which shows that spatial context contributes useful information.

Table X lists the top 10 features with their importance scores.

Environmental features rank in the lower part of the importance list, suggesting that weather and road-condition variation in this dataset adds less predictive separation than temporal, vehicle, and driver-related variables.

Learning curve. Fig. 10 shows a clear case of overfitting: training accuracy reaches 1.0 while validation accuracy plateaus near 0.85.

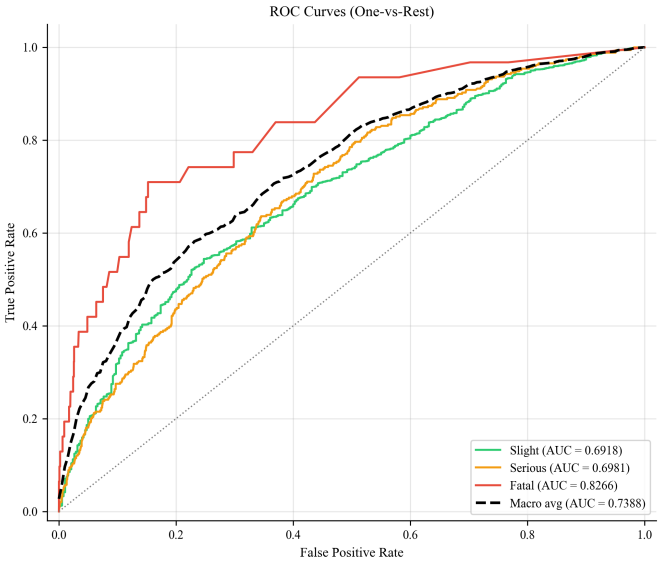

Fig. 8: ROC curves (one-vs-rest) for three severity classes. Macro-average AUC = 0.7388.

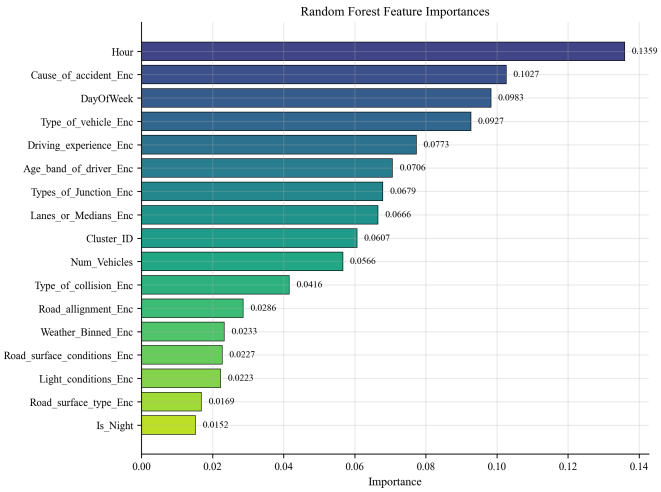

Fig. 9: Random Forest feature importances ranked by contribution. Hour of day is the strongest predictor.

TABLE X: Top 10 Feature Importances

Rank	Feature	Importance
1	Hour	0.1359
2	Cause_of_accident	0.1027
3	DayOfWeek	0.0983
4	Type_of_vehicle	0.0927
5	Driving_experience	0.0773
6	Age_band_of_driver	0.0706
7	Types_of_Junction	0.0679
8	Lanes_or_Medians	0.0666
9	Cluster_ID	0.0607
10	Num_Vehicles	0.0566

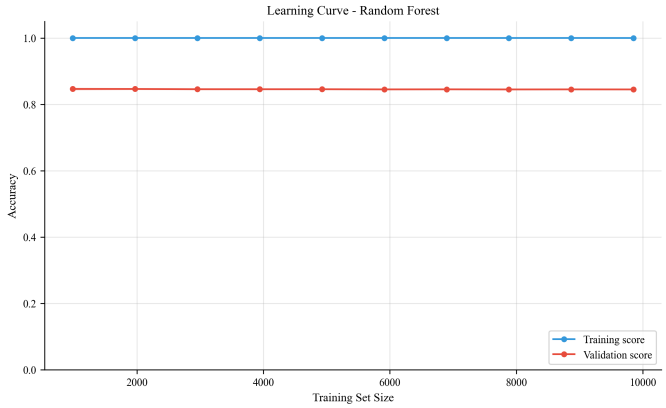

Fig. 10: Learning curve for the Random Forest classifier. The persistent gap between training and validation scores indicates overfitting.

Cross-validation. Table XI shows stable accuracy across five folds (0.8476 ± 0.0010), while macro F1 varies more (0.3251 ± 0.0112), reflecting unstable minority-class predictions.

TABLE XI: Five-fold Stratified Cross-validation Results

Fold	Accuracy	F1 (macro)	Precision	Recall
1	0.8466	0.3150	0.4908	0.3378
2	0.8486	0.3208	0.5791	0.3408
3	0.8490	0.3417	0.9125	0.3516
4	0.8473	0.3133	0.6157	0.3372
5	0.8465	0.3345	0.6575	0.3479
Mean	0.8476	0.3251	0.6511	0.3430
Std	0.0010	0.0112	0.1418	0.0057

D. *Comparative analysis*

To place the Random Forest results in context, five additional classifiers were trained on the same split and feature set. Table XII shows that the Random Forest model has the best accuracy. In contrast, the Decision Tree model achieves the highest macro F1-score, which indicates that its predictions are slightly more balanced.

TABLE XII: Comparative Performance of Classification Models

Model	Acc.	Prec.	Rec.	F1	κ
Decision Tree	0.737	0.352	0.353	0.352	0.068
Random Forest	0.847	0.491	0.338	0.315	0.020
Naive Bayes	0.843	0.359	0.335	0.310	0.006
KNN	0.839	0.312	0.332	0.308	-0.006
SVM (RBF)	0.500	0.355	0.408	0.306	0.046
Logistic Reg.	0.436	0.360	0.446	0.291	0.043

All models indicate low Cohen's Kappa values indicating weak separation of the minority class, with features available to better capture temporal and spatial patterns as opposed to crash dynamics.

E. Accident Risk Index results

The ARI was computed for all 9 clusters using the formula in (1) with derived weights $W_1 = 0.534$, $W_2 = 0.401$, $W_3 = 0.065$. Table XIII shows the component scores and final ARI for each cluster.

TABLE XIII: Accident Risk Index Results per Cluster

ID	S_{sev}	S_{dens}	S_{env}	ARI	Tier
0	0.428	1.000	0.15	0.639	Severe
1	0.116	0.608	0.15	0.315	Moderate
6	1.000	0.011	0.15	0.548	Severe
7	0.749	0.037	0.15	0.424	Moderate
4	0.375	0.055	0.15	0.232	Low
2	0.346	0.056	0.15	0.217	Low
3	0.110	0.078	0.15	0.100	Low
5	0.000	0.180	0.15	0.082	Low
8	0.043	0.071	0.15	0.061	Low

ARI scores range from 0.061 (Cluster 8, Chennai area) to 0.639 (Cluster 0, Pune/Mumbai area). No cluster reaches the Critical tier. Cluster 0 ranks highest because it combines the greatest density with moderate severity, while Cluster 6 reaches the Severe tier mainly because of its high severity score despite a small incident count.

Fig. 11 shows the risk map, where each cluster centroid is plotted with a color corresponding to its ARI score and a bubble size proportional to incident count.

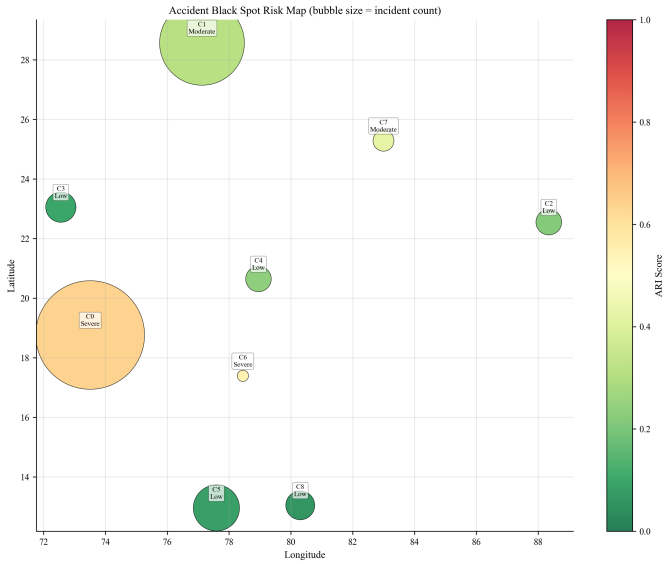

Fig. 11: Accident black spot risk map. Circle size is proportional to incident count; color indicates ARI score (green = low, red = high).

Fig. 12 shows the severity composition across clusters, illustrating how severity varies geographically.

F. Discussion

The system has a clear strength: DBSCAN finds geographically coherent clusters without pre-defining the number of regions, and the ARI turns those outputs into a priority list that

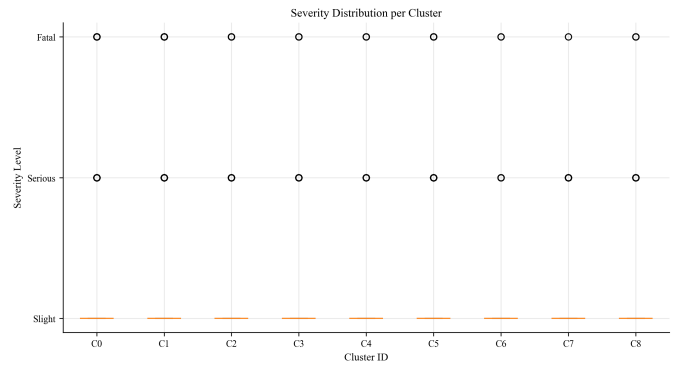

Fig. 12: Severity composition by cluster, showing the proportion of Slight, Serious, and Fatal injuries within each cluster.

is easier to act on than raw counts alone. Including Cluster_ID in the classifier also adds measurable spatial context.

The main limitation of this approach is class imbalance. With 84.56% of records belonging to the Slight Injury class, the classifier will already tend to select Slight for most inputs. Balanced class weighting helps to alleviate this but, on its own, is not sufficient. This limitation is exacerbated by several other factors:

1) The features in the dataset describe accident circumstances (weather, road type, junction type) but not crash dynamics (speed, impact angle, vehicle deformation). Severity is largely determined by crash dynamics, which are not available in police-report-style data.
2) The dataset includes only 158 records of fatal injuries, which is a very small 1.28% of the total. This small number makes it difficult for a tree-based ensemble model to find consistent patterns, even if class weighting is used.
3) The geocoding step maps categorical area types to city-level coordinates with jitter. This introduces spatial noise that may dilute the spatial signal available to the classifier.

Despite that shortcoming, the solution gives access to a full pipeline from raw data to a risk-tiered map, and the Flask API ensures the workflow is accessible from external applications.

VI. Conclusion and Future Work

This paper presented an integrated accident hotspot prediction system that combines DBSCAN spatial clustering, Random Forest classification, and a data-driven ARI.Using 12,316 Indian road accident records, DBSCAN found nine clusters. The Random Forest model then achieved a test accuracy of 84.66%. The hour of the day, the cause of the accident, and the day of the week were the most important factors in predicting accident outcomes.The ARI then separated the clusters into two Severe, two Moderate, and five Low-risk groups.

At the same time, the classifier remains weak on minority classes. Cohen's Kappa of 0.0198 and a macro F1-score of 0.3149 show that Serious and Fatal injuries are not being separated reliably.

Several directions could improve future versions of the system:

1) **Oversampling techniques.** SMOTE or ADASYN could give the classifier more minority-class signal during training.
2) **Richer spatial data.** Substituting geocoded area labels with real GPS coordinates would most probably enhance clustering quality and Cluster_ID utility.
3) **Alternative models.**Models that incorporate spatial structure might be better at capturing interactions that tree-based methods could miss.
4) **Real-time integration.** Live traffic and weather feeds could transform the current workflow into an early-warning system.
5) **Additional features.** Vehicle speed, hospital proximity, and emergency response time could improve severity prediction.
6) **Binary classification reformulation.** Collapsing the three severity classes into two may produce a more practical classifier by reducing the imbalance problem.

The source code, trained models, and API are publicly available at the project repository.

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